Introduction to Computing and the Computing Profession

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Learning Outcomes



At the end of the lecture, students should be able to:

1. define computing and related disciplines
2. distinguish between Computer Science, Information Technology, Information Systems, and Software Engineering
3. describe the major components of a computing environment
4. explain the historical development of computing
5. identify major categories of computers
6. discuss major applications of computing
7. identify common careers in computing
8. explain the role of computing professionals in society

What is Computing?

Computing refers to the use of computers, software, data, networks, algorithms, and related technologies to process information and solve problems.

Computing includes:

- designing computer systems
- writing software
- storing and analysing data
- communicating through networks
- developing algorithms
- securing information
- automating tasks



Why Computing Matters



Mobile money



Online learning



Social media



Banking



Healthcare



Transport



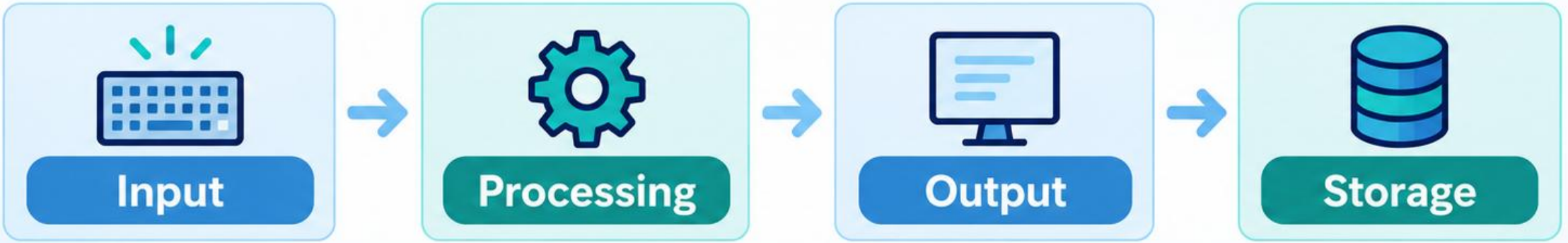
Government services

What is a Computer?

A computer is an electronic device that accepts data as input, processes the data according to instructions, stores data and information, and produces output.



A basic computer system performs four major functions:



Input–Process–Output–Storage Example

Example: ATM Transaction

1 Input



Card, PIN, withdrawal amount

2 Processing



Verify account and balance

3 Output



Cash and receipt

4 Storage



Transaction record

Components of a Computing Environment



Hardware



Software



Data



People



Procedures



Communication and Networks

Hardware

Hardware refers to the physical components of a computer system.



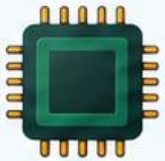
Examples



keyboard



monitor



processor



mouse



printer



hard disk



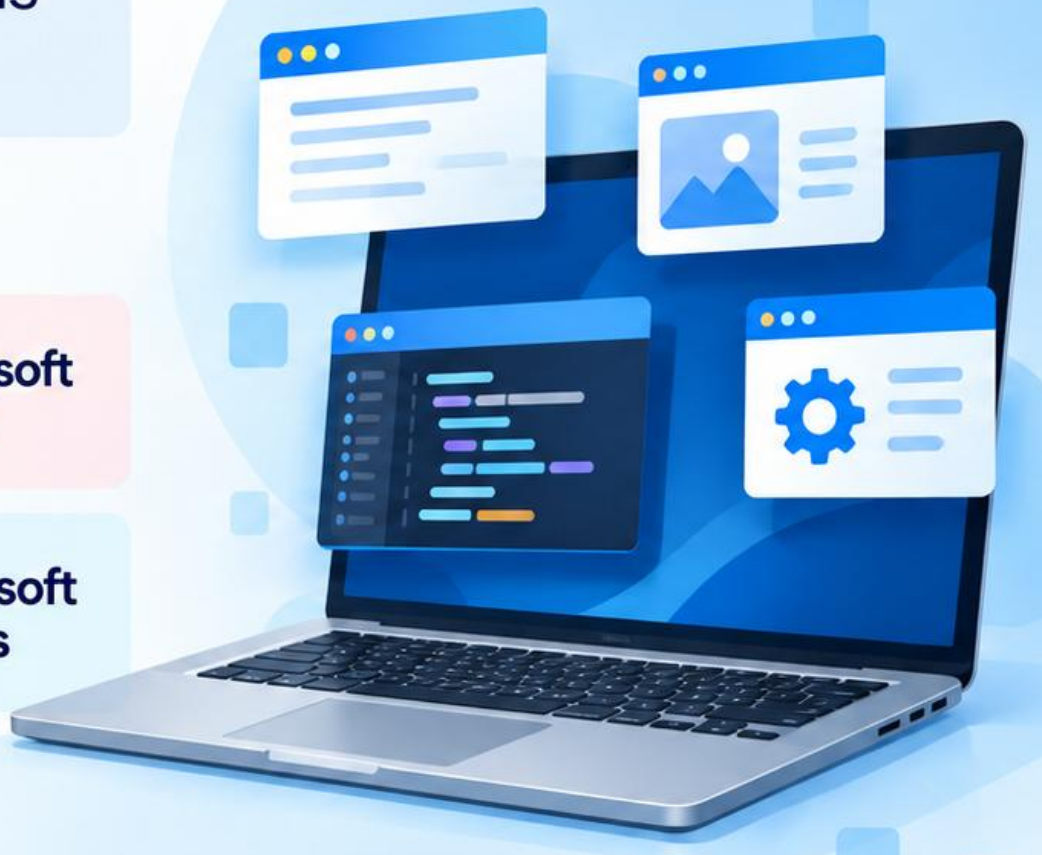
network switch

Software

Software consists of programs and instructions that tell computer hardware what to do.

Examples:

- Microsoft Windows
- Linux
- Microsoft Word
- Google Chrome
- WhatsApp
- Microsoft Teams
- Antivirus software



Data

Data refers to raw facts that can be processed by a computer.



Examples

- student registration number
- date of birth
- name
- photograph
- examination score
- transaction amount

People

People interact with computing systems.

Examples

- students
- lecturers
- software developers
- system administrators
- customers
- managers
- database administrators



Procedures

Procedures are rules or steps that specify how a computing system should be used.

Examples

- steps for registering courses
- password-reset procedure
- procedure for backing up files
- procedure for processing student results



Communication and Networks

Computing systems often communicate through networks.

Examples

- Wi-Fi
- campus network
- mobile data network
- Internet
- Bluetooth



How the Components Work Together

Example: University LMS

A computing system is more than just hardware and software!



Hardware

Laptop, phone, servers

Software

LMS application





Data

Student records, course materials, grades

University LMS

Brings all the components together to support teaching and learning

People

Students, lecturers, administrators





Procedures

Login, submission, grading

Networks

Internet/Wi-Fi



Major Areas of Computing



Computer Science



Information Technology



Information Systems



Software Engineering

Computer Science

Computer Science focuses on the principles and theories behind computation and the design of computational solutions.



Major areas include:

Algorithms
Methods for solving problems.

Programming
Writing instructions for computers.

Data Structures
Organizing and managing data.

Artificial Intelligence
Building systems that think and learn.

Computer Graphics
Creating and manipulating images.

Operating Systems
Managing computer hardware and software.

Computer Architecture
Designing computer hardware.

Theory of Computation
Understanding what can be computed.

Machine Learning
Learning from data to make predictions.

Information Technology



Information Technology focuses on the practical use, deployment, management, and support of computing technologies.

Common areas include:

- Networking
- System Administration
- Cybersecurity
- Cloud Computing
- IT Support
- Infrastructure Management
- Database Administration

Technology for a Better Tomorrow

Information Systems



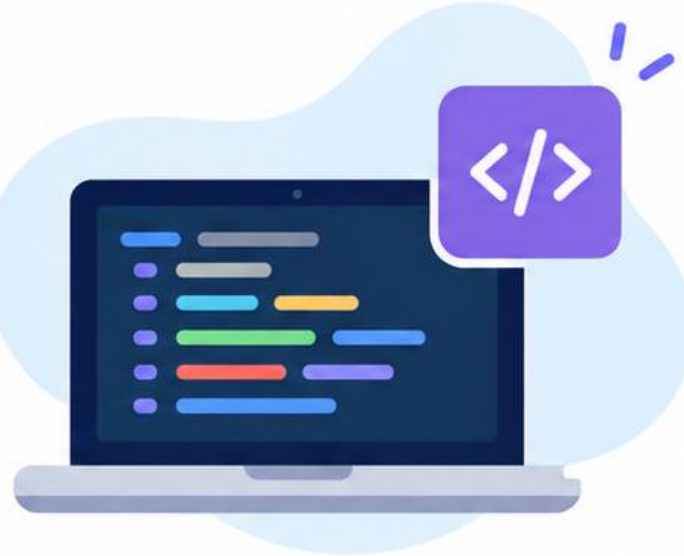
Information Systems focuses on how organisations use people, processes, data, and technology to support operations and decision-making.

Examples include:

- **Student information systems**
- **Hospital management systems**
- **Banking systems**
- **Payroll systems**
- **Library management systems**

People + Processes + Data + Technology for Better Decisions

Software Engineering



Software Engineering is concerned with the systematic design, development, testing, deployment, and maintenance of software systems.

It emphasises:

Requirements Analysis
Understand what the system should do

Software Design
Plan the structure and solution

Coding
Write the software

Testing
Check for errors and ensure quality

Documentation
Create and maintain system documentation

Maintenance
Fix issues and make improvements

Project Management
Plan, coordinate and deliver projects

Major Areas of Computing



Computer Science



Information Technology



Information Systems



Software Engineering

Related Areas of Computing



Cybersecurity

Focuses on protecting computers, networks, systems, and data from unauthorised access or attack.



Data Science

Focuses on extracting useful ‘insights from data.



Artificial Intelligence

Focuses on developing systems capable of performing tasks that normally require human intelligence.

Identify the Most Relevant Discipline

Which computing discipline is most closely related to each activity?

1. 

   Designing a faster search algorithm → ?

2. 

   Managing university network infrastructure → ?

3. 

   Developing a payroll system → ?

4. 

   Building and testing a mobile application → ?

5. 

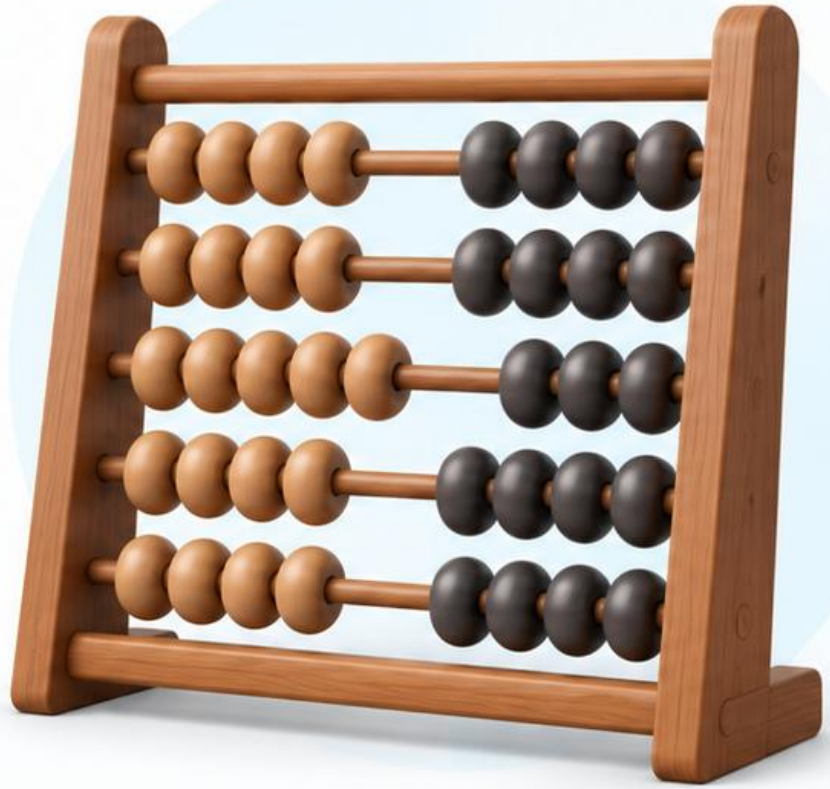
   Detecting suspicious network activity → ?

Evolution of Computing

Early Computing Devices

Abacus

One of the earliest devices used for arithmetic calculations.
It was used for counting, addition and subtraction.



Evolution of Computing

PEOPLE
IDEAS
MACHINES
A BRIGHTER TOMORROW

Mechanical Calculators

EARLY INNOVATIONS THAT LAID
THE FOUNDATION FOR MODERN COMPUTING

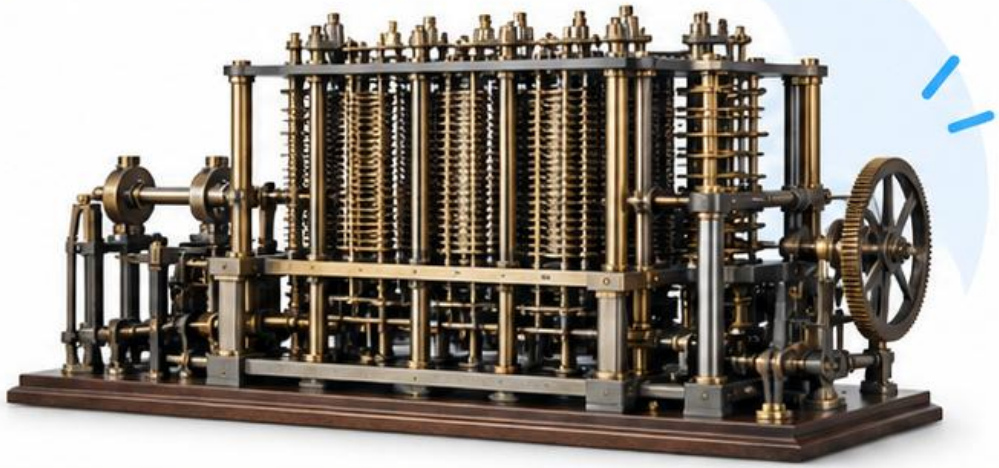
01 Pascaline

Developed by Blaise Pascal in the 17th century. It could perform basic arithmetic.



02 Analytical Engine

Designed by Charles Babbage.
Important concepts included input, processing, memory and output.



Generations of Computers

1 First Generation

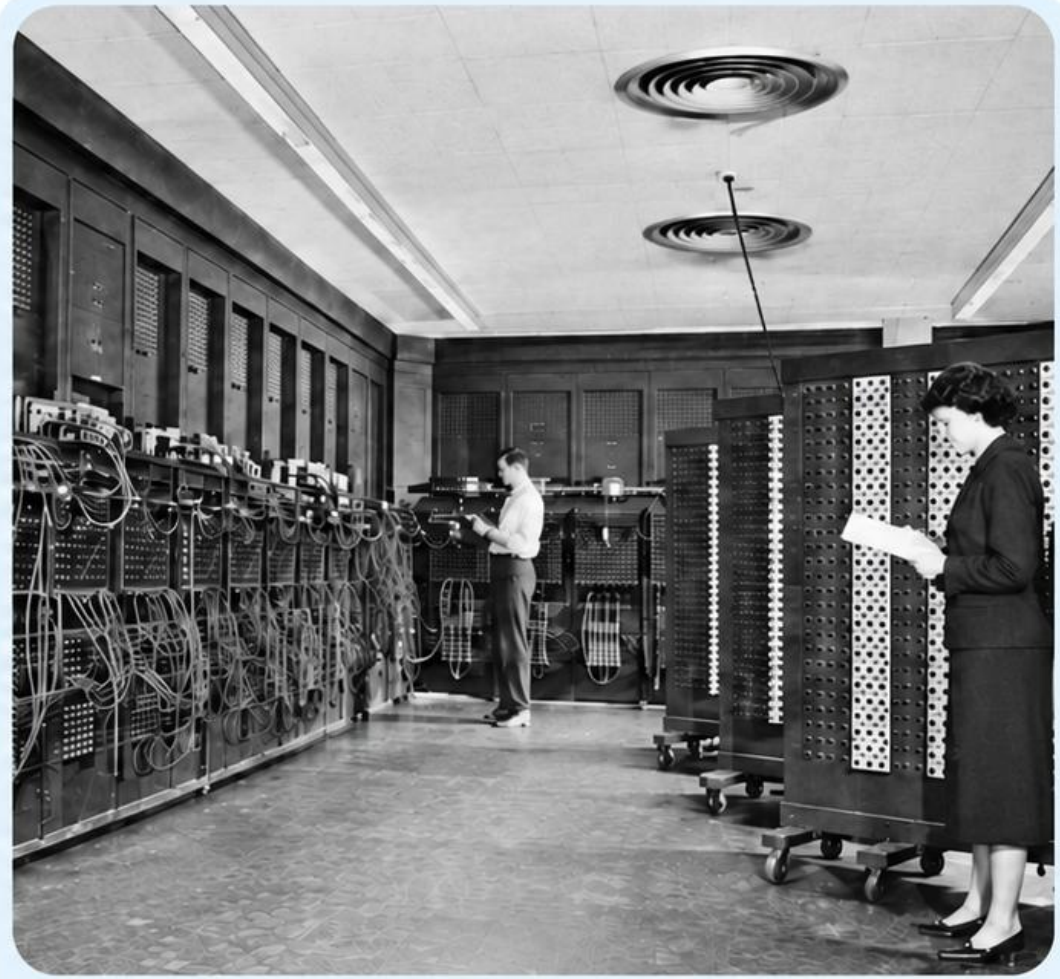
The beginning of modern computing.

Approximate period
1940s–1950s.

Main technology
Vacuum tubes.

Characteristics

- Very large
- Generated significant heat
- Expensive
- Consumed large amounts of electricity



ENIAC (1940s)

Generations of Computers

2 Second Generation

Used transistors instead of vacuum tubes, leading to smaller, faster, and more reliable computers.

Main technology

Transistors.

Advantages over vacuum tubes

- Smaller
- Faster
- More reliable
- Consumed less energy



IBM 1401 (1959)

Year introduced
1959

Manufacturer
IBM

Generation
Second Generation (Transistors)

Main use
Business data processing

Key feature
Reliable, widely used, and more affordable than earlier computers



Generations of Computers

3 Third Generation

Main technology
Integrated circuits.

Multiple electronic components were placed on a single chip.

Benefits

- improved speed
- greater reliability
- smaller size



Generations of Computers

4 Fourth Generation

Main technology
Microprocessors.

The CPU could be placed on a single chip.

This development contributed to:

- personal computers
- laptops
- smartphones
- embedded computing systems



Generations of Computers

Fifth Generation and Modern Computing

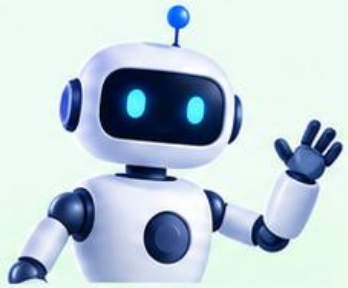
Modern computing is characterised by developments such as:



artificial intelligence



cloud computing



robotics



machine learning



Internet of Things



mobile computing



quantum computing research

Transition from First to Fifth Generation Computers

How computer technology evolved over time

1 First Generation

1940s–1950s



Vacuum tubes

Very large, expensive, produced much heat

2 Second Generation

1950s–1960s



Transistors

Smaller, faster, more reliable

3 Third Generation

1960s–1970s



Integrated Circuits (ICs)

More efficient and widely used

4 Fourth Generation

1970s–present



Microprocessors

Personal computers became common

5 Fifth Generation

Present and beyond



Artificial Intelligence

Smart systems, automation, learning machines

Categories of Computers



Personal Computers



Workstations



Servers



Mainframe Computers



Supercomputers



Mobile Devices



Embedded Systems

Personal Computers

Designed primarily for individual use.

Examples

- desktop computers
- laptops



Workstations

High-performance computers typically used for professional or technical applications.

Examples:

- engineering
- scientific computing
- graphics
- AI development



Servers

A server provides services or resources to other computers.

Examples:

- Web server
- Database server
- File server
- Email server



Mainframe Computers

Powerful computers designed to process very large volumes of transactions and support many users simultaneously.

Used by:

- banks
- governments
- insurance companies
- large organisations



Supercomputers

Extremely powerful systems used for complex scientific calculations.

Applications include:

- climate modelling
- weather forecasting
- space research
- scientific simulation
- artificial intelligence



Mobile Devices

Mobile devices are portable computing devices designed for use on the go.

Examples:



smartphones



tablets



smartwatches



Embedded Systems

An **embedded system** is a computer built into another device to perform a specific task.

Examples:



Washing machine



Microwave



Vehicle



Traffic light



Digital television



Smart meter



ATM

Choosing the Right Type of Computer

	Activity / Task		Most Suitable Type of Computer
1		Weather forecasting →	?
2		Bank transaction processing →	?
3		Hosting a university website →	?
4		Student writing an assignment →	?
5		Washing-machine controller →	?
6		Smartphone banking →	?

Applications of Computing



Education



Banking and Finance



Healthcare



Agriculture



Transportation



Government



Entertainment

APPLICATIONS OF COMPUTING

Education

Computing in education helps improve teaching and learning by providing digital tools and resources.

Examples:

- Learning Management Systems
- Online examinations
- Virtual classrooms
- Student-information systems
- Digital libraries
- AI learning assistants



Learn • Create • Achieve

Banking and Finance

PEOPLE
IDEAS
FINANCIAL
OPPORTUNITIES
BRIGHTER
TOMORROWS

Applications:

- ATMs
- mobile banking
- online banking
- fraud detection
- electronic payments



ATMs


mobile banking


online banking


fraud detection


electronic payments



APPLICATIONS OF COMPUTING

Healthcare

Computing in healthcare improves patient care by enabling efficient management, diagnosis and treatment.

Applications:

- Electronic medical records
- Telemedicine
- Diagnostic imaging
- Appointment systems
- AI-assisted diagnosis



Agriculture

Applications:

- weather monitoring
- crop-management systems
- drone-based farm monitoring
- digital marketplaces
- automated irrigation



Transportation

Computing in transportation makes travel safer, faster and more efficient.

Applications:



GPS navigation



ride-hailing applications



traffic-management systems



airline reservation systems



Government

Applications:

- tax systems
- passport services
- digital identity
- online application systems
- public-record management



Entertainment

Examples:



Video streaming



Gaming



Music streaming



Social media



Digital content creation

Computing Careers



Skills of a Computing Professional

Technical Skills

Specialised skills for working with computing technologies.

1. **Programming**



2. **Networking**



3. **Databases**



4. **Systems administration**



5. **Cybersecurity**





Professional Skills

Broader skills that help you work effectively with people and in the workplace.

1. **Communication**
2. **Teamwork**
3. **Problem solving**
4. **Ethics**
5. **Adaptability**
6. **Continuous learning**

A successful computing professional combines both **technical skills** and **professional skills.**

Computing in Everyday Life

What computing devices or systems did you use today before coming to class?



Computing in Everyday Life



What computing devices or systems did you use today before coming to class?



smartphone alarm



mobile money



social media



university portal



GPS



ATM



Google Search



email



WhatsApp



transport application

Identify the Computing System



Consider the following systems:

- ATM system
- University LMS
- Mobile money system
- Hospital information system
- Online shopping system
- Ride-hailing application

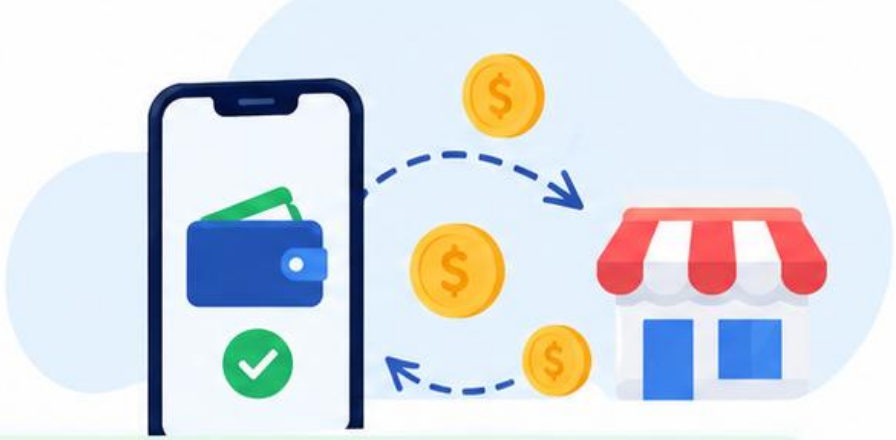


For each system identify:

- users
- hardware
- software
- data
- network requirements
- input
- processing
- output

Mobile Money

Sending money. Paying for services. A more connected world.





Users

- customer
- merchant
- mobile money agent
- service provider



Hardware

- smartphone
- server
- network equipment



Software

- mobile money application
- transaction-processing software



Data

- phone number
- account balance
- transaction amount
- PIN

Mobile Money

From input to a completed transaction



Fast
Secure
Convenient

Input

User provides transaction details.

Amount
How much to send?

Recipient number
Who to send to?

PIN
Verify identity

Processing

System performs checks and completes the transfer.

Verify account
Check account exists

Verify PIN
Authenticate user

Check available balance
Ensure sufficient funds

Transfer funds
Complete the transaction

Output

User receives confirmation of the transaction.



Transaction confirmation
Notification or receipt

Mobile Money: Examples for Each Category

Users

- Customer
- Merchant
- Mobile money agent
- Service provider

Hardware

- Smartphone
- Agent phone or POS device
- Server
- Network equipment

Software

- Mobile money app
- USSD menu
- Transaction-processing software
- Fraud-detection software

Data

- Phone number
- Wallet balance
- Transaction amount
- PIN
- Transaction record

Network Requirements

- Mobile network
- Internet connectivity
- Secure payment gateway

Input

- Amount
- Recipient number
- PIN

Processing

- Verify user
- Verify PIN
- Check balance
- Transfer funds
- Update records

Output

- Transaction confirmation
- Updated balance
- Receipt or reference number